

Comprehensive Internship Project Report:

Unemployment Analysis in India (Pre & Post COVID-19 Impact)

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Internship Program: Oasis Infobyte Data Science Internship

Task Track: Task 2 Unemployment Analysis with Python

Domain: Exploratory Data Analysis (EDA), Time-Series Analytics & Macroeconomic Modeling

Associated Notebook: Unemployment_Analysis.ipynb

Date: September 2026

Executive Summary

For my second internship task, I performed an extensive Exploratory Data Analysis (EDA) on unemployment data in India to analyze regional and temporal trends, evaluate rural versus urban workforce dynamics, and measure the severe economic shock caused by the COVID-19 pandemic and the subsequent March 2020 national lockdown.

Economic dataset analysis requires meticulous data wrangling, cleaning, and time-series aggregation to convert raw survey numbers into clear, meaningful socio-economic stories.

During this project, I executed the following workflow:

1. Multi-Dataset Ingestion & Metadata Inspection
2. Advanced Data Hygiene: Header Stripping, Missing Row Audit & String Trimming
3. Strict Date Parsing & Temporal Feature Engineering (`Year`, `Month`, `COVID\_Period`)
4. National & Area-Wise (Rural vs. Urban) Descriptive Analytics
5. Geographic Ranking Analysis (Top & Bottom Unemployment States)
6. Time-Series Line Trajectory Modeling & Lockdown Peak Annotations
7. Correlation Heatmap Analysis of Macroeconomic Indicators
8. Pre-COVID Baseline vs. Post-COVID Lockdown Comparative Impact Analysis
9. Socio-Economic Synthesis & Policy Takeaways

The analysis revealed that following the nationwide lockdown on March 24, 2020, national unemployment surged from a pre-COVID baseline of **9.47%** to an unprecedented peak of **23.60%** in April 2020. Urban areas were hit substantially harder (**25.84%** peak) than rural areas (**21.48%** peak) due to service sector halts and urban informal labor dislocation.

1. Problem Statement & Dataset Background

1.1 Economic & Analytical Context

Unemployment rate is a crucial macroeconomic indicator measuring the percentage of the active labor force that is unemployed and actively seeking employment:

$$\text{Unemployment Rate (\%)} = \frac{\text{Unemployed Individuals}}{\text{Total Labour Force}} \times 100$$

The arrival of the COVID-19 pandemic in early 2020 prompted the Government of India to declare a strict 21-day nationwide lockdown on March 24, 2020, which was subsequently extended in multiple phases through May 2020. This sudden economic halt disrupted transportation, manufacturing, retail, and informal services.

1.2 Datasets Ingested

I analyzed two complementary datasets sourced from Centre for Monitoring Indian Economy (CMIE) survey data:

10. `Unemployment in India.csv` (740 non-null records): Contains monthly unemployment metrics categorized by **Area (Rural vs. Urban)** from May 2019 to June 2020.
11. `Unemployment\_Rate\_upto\_11\_2020.csv` (267 non-null records): Contains state-level monthly metrics with **Geographic Coordinates (Latitude/Longitude)** and **Macro-Region (Zone)** classifications (North, South, East, West, North East, Central) from January 2020 to October 2020.

2. Data Preprocessing & Hygiene Audit (My Step-by-Step Approach)

Real-world economic data often suffers from formatting inconsistencies. I performed a rigorous multi-step cleaning pipeline to ensure analytical integrity:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
```

```
warnings.filterwarnings('ignore')

# 1. Load raw CSV files
df_area = pd.read_csv('Unemployment in India.csv')
df_rate = pd.read_csv('Unemployment_Rate_upto_11_2020.csv')

# 2. Header Normalization: Strip trailing whitespaces
df_area.columns = df_area.columns.str.strip()
df_rate.columns = df_rate.columns.str.strip()

# 3. Drop null padding rows
df_area = df_area.dropna().reset_index(drop=True)
df_rate = df_rate.dropna().reset_index(drop=True)

# 4. Clean text column values
for col in ['Region', 'Area', 'Frequency']:
    if col in df_area.columns:
        df_area[col] = df_area[col].astype(str).str.strip()

for col in ['Region', 'Frequency', 'Region.1']:
    if col in df_rate.columns:
        df_rate[col] = df_rate[col].astype(str).str.strip()

# Rename Region.1 to Macro_Region for clarity
if 'Region.1' in df_rate.columns:
    df_rate.rename(columns={'Region.1': 'Macro_Region'}, inplace=True)
```

2.1 Resolving the Date String Formatting Bug

When converting the `Date` column to `datetime64[ns]`, I encountered a formatting error: raw date values contained leading whitespace (e.g. `" 31-05-2019")`).

To resolve this, I implemented string trimming before date conversion:

```
df_area['Date'] = df_area['Date'].astype(str).str.strip()
df_rate['Date'] = df_rate['Date'].astype(str).str.strip()

df_area['Date'] = pd.to_datetime(df_area['Date'], format='%d-%m-%Y')
df_rate['Date'] = pd.to_datetime(df_rate['Date'], format='%d-%m-%Y')
```

2.2 Feature Engineering: Temporal & Period Feature Categorization

To facilitate timeline comparisons, I extracted additional temporal columns:

```
# Feature Extraction for df_area
df_area['Year'] = df_area['Date'].dt.year
```

```
df_area['Month'] = df_area['Date'].dt.month
df_area['Month_Name'] = df_area['Date'].dt.strftime('%b %Y')

# Define COVID Lockdown Era Split (Pre-COVID = < March 2020, Post-COVID = >= March 2020)
df_area['COVID_Period'] = np.where(
    df_area['Date'] < '2020-03-01',
    'Pre-COVID (May 2019 - Feb 2020)',
    'COVID Lockdown (Mar 2020 - Jun 2020)'
)
```

3. Exploratory Data Analysis & Descriptive Statistics

I calculated descriptive summary statistics overall and grouped by Area (Rural vs. Urban).

Overall Summary Statistics Table (`Unemployment in India.csv`)

Metric Indicator	Count	Mean	Std Dev	Min	25% IQR	Median	75% IQR	Max
Estimated Unemployment Rate (%)	740	11.788%	10.738%	0.000%	4.590%	8.350%	15.888%	75.850%
Estimated Employed	740	**13.96M**	13.37M	49.42K	2.79M	9.87M	21.19M	60.40M
Estimated Labour Participation Rate (%)	740	**42.630%**	8.111%	13.330%	37.180%	41.160%	45.880%	72.570%

Summary Statistics Table Grouped by Area (Rural vs Urban)

Area Type	Sample Count	Mean Unemployment Rate (%)	Mean Employed Count	Mean Labour Participation Rate (%)
Rural	359	10.323%	19.03 Million	44.464%

Urban

381

13.169%

9.19 Million

40.901%

4. Regional Analysis: Top & Bottom States Analysis

I analyzed state-level average unemployment rates to identify geographic hotspots:

4.1 Top 10 Highest Unemployment States / Union Territories

12. **Puducherry: 20.83%** mean unemployment
13. **Tripura: 20.81%** mean unemployment
14. **Haryana: 19.43%** mean unemployment
15. **Jharkhand: 19.04%** mean unemployment
16. **Bihar: 18.91%** mean unemployment
17. **Jammu & Kashmir: 16.19%** mean unemployment
18. **Delhi: 15.76%** mean unemployment
19. **Goa: 12.58%** mean unemployment
20. **Himachal Pradesh: 12.04%** mean unemployment
21. **Tamil Nadu: 12.02%** mean unemployment

4.2 Top 10 Lowest Unemployment States / Union Territories

22. **Meghalaya: 4.80%** mean unemployment
23. **Odisha: 5.65%** mean unemployment
24. **Assam: 6.43%** mean unemployment
25. **Gujarat: 6.67%** mean unemployment
26. **Telangana: 6.68%** mean unemployment
27. **Karnataka: 6.68%** mean unemployment
28. **West Bengal: 7.88%** mean unemployment
29. **Chhattisgarh: 7.89%** mean unemployment
30. **Madhya Pradesh: 7.91%** mean unemployment
31. **Maharashtra: 7.98%** mean unemployment

5. Temporal & Time-Series Line Trajectory Analysis

5.1 National Monthly Unemployment Rate Trend (May 2019 – June 2020)

I plotted the national mean unemployment rate over time and overlaid a semi-transparent red highlight (`plt.axvspan``) representing the peak national lockdown period (March 24 – May 31, 2020).

Monthly National Average Unemployment Rate Trajectory:

- **May 2019:** 8.83%
- **June 2019:** 9.30%
- **July 2019:** 9.04%
- **August 2019:** 9.58%
- **September 2019:** 9.05%
- **October 2019:** 9.94%
- **November 2019:** 9.61%
- **December 2019:** 9.49%
- **January 2020:** 9.94%
- **February 2020:** 9.96%
- **March 2020 (Lockdown Begins):** 10.70%
- **April 2020 (Peak Lockdown Shock):** **23.60%** (Surge of +13.64 percentage points in a single month!)
- **May 2020 (Continued Lockdown):** **20.88%**
- **June 2020 (Unlock Phase 1 Begins):** **10.20%** (Rapid V-shaped recovery)

5.2 Time-Series Trajectory for Major Regions

I selected 7 representative regions (**Puducherry, Haryana, Jharkhand, Delhi, Maharashtra, Tamil Nadu, Andhra Pradesh**) and plotted multi-line time-series trends:

- **Puducherry:** Recorded the single highest unemployment spike in the dataset, soaring from 7.58% in February 2020 to 75.85% in April 2020!
- **Jharkhand:** Surged from 11.60% in Feb 2020 to 59.20% in April 2020.
- **Tamil Nadu:** Surged from 6.44% in Feb 2020 to 49.83% in April 2020.
- **Delhi:** Surged from 14.84% in Feb 2020 to 42.47% in April 2020.

6. Correlation Heatmap Analysis

I computed the Pearson correlation matrix across the three continuous macroeconomic features:

Pearson Correlation Matrix Table

Feature Indicator	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)
Estimated Unemployment Rate (%)	1.000	-0.245	-0.086
Estimated Employed	-0.245	1.000	+0.012
Estimated Labour Participation Rate (%)	-0.086	+0.012	1.000

Analytical Insights:

32. **Unemployment Rate vs. Employed Count (\$r = -0.245\$)**: Moderate negative correlation. As expected, when workforce employment contracts, the unemployment rate rises.
33. **Unemployment Rate vs. Labour Participation Rate (\$r = -0.086\$)**: Weak negative correlation. This reveals that despite massive job losses during lockdown, labor market participation did not collapse proportionally, as dislocated workers actively sought new employment.

7. Pre-COVID vs. Post-COVID Pandemic Impact Analysis

I partitioned the dataset to compare the baseline pre-COVID period against the lockdown period:

State-Wise Pre-COVID vs. Post-COVID Impact Table (Top 10 Affected States)

State / Region	Pre-COVID Mean (%) (May 19 - Feb 20)	Post-COVID Mean (%) (Mar 20 - Jun 20)	Absolute Increase (% Points)	Percentage Growth (%)
Puducherry	8.12%	38.45%	+30.33%	+373.5%
Jharkhand	12.85%	34.52%	+21.67%	+168.6%
Tamil Nadu	4.21%	25.28%	+21.07%	+500.5%
Haryana	16.54%	31.10%	+14.56%	+88.0%

Bihar	15.42%	27.63%	+12.21%	+79.2%
Delhi	13.08%	22.45%	+9.37%	+71.6%
Andhra Pradesh	5.45%	14.28%	+8.83%	+162.0%
Maharashtra	5.18%	14.98%	+9.80%	+189.2%
Punjab	10.82%	19.45%	+8.63%	+79.8%
West Bengal	6.15%	12.20%	+6.05%	+98.4%

8. Key Takeaways & Internship Reflections

34. **Handling Raw Formatting & Date Strings:** I gained practical experience resolving pandas datetime parsing errors caused by hidden leading whitespace in string values.
35. **Contextualizing Time-Series Visualizations:** I learned how adding date annotations (`plt.annotate``) and shaded spans (`plt.axvspan``) transforms a standard line chart into a compelling story of an external economic event.
36. **Urban Vulnerability in Pandemics:** The analysis proved that urban centers experienced steeper job losses during lockdown due to service sector dependency, whereas rural areas showed faster economic recovery due to agriculture and MGNREGA employment programs.